MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS
SEMICUSTOM (GATE ARRAY) DEVICES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, silicon gate CMOS semicustom (gate array) devices. Two product assurance classes (B and S) and a choice of case outlines and lead finishes are provided for each type, and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device types. The device types (total number of usable gates) and circuit organization shall be as identified in the specific altered item drawing (AID) and as follows:

Device type	Circuit
01	≤1,000 gate array
02	≤2,000 gate array
03	≤3,000 gate array
04	≤4,000 gate array
05	≤5,000 gate array
06	≤6,000 gate array
07	≤7,000 gate array
08	≤8,000 gate array
09	≤9,000 gate array
10	≤10,000 gate array
11	≤11,000 gate array
12	≤12,000 gate array
13	≤15,000 gate array
14	≤25,000 gate array
15	≤35,000 gate array
16	≤45,000 gate array
17	≤55,000 gate array

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines and letters shall be in accordance with MIL-M-38510, appendix C.

Outline letter	Case outline	Manufacturers (see table V)
N	C-G6 (84-terminal, leaded square chip carrier package with unformed leads)	B
T	P-BJ (208-pin, square pin grid array)	B
U	P-AC (84-pin, square pin grid array)	A
X	C-7 (68-terminal), square chip carrier package	C
Y	C-8 (84-terminal), square chip carrier package	D
Z	P-AG (180-pin, square pin grid array)	B

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings.

Supply voltage range	-0.3 V dc to +7.0 V dc
DC input, dc output voltage ranges (V_{IN}, V_{OUT})	-0.3 V dc to (V_{DD} +0.3 V dc)
Storage temperature range	-65°C to +150°C
Maximum junction temperature (T_J)	+175°C
Thermal resistance, junction-to-case (θ_{JC})	(See MIL-M-38510, appendix C)

1.4 Recommended operating conditions.

Supply voltage (V_{DD})	4.5 V dc to 5.5 V dc
DC input voltage	0 V dc to V_{DD}.
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-976 - Certification Requirements for JAN Microcircuits.

(Copies of the specification and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

JEDEC Standard No. 12 - Standard for Gate Array Benchmark Set.
JEDEC Standard No. 12-3 - CMOS Gate Array Macrocell Standard.

(Applications for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained. The altered item drawing (AID) that forms a part of this specification shall not take precedence over the requirements of this specification. 1/

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Altered item drawing (AID) requirements. The following items must be provided to the device manufacturer by the customer as part of an altered item drawing.

3.2.1 Terminal connections and pin assignments.

3.2.2 Package type selected from manufacturer's offering and in accordance with MIL-M-38510, appendix C (see 1.2.3).

3.2.3 Functional block diagram.

3.2.4 Functional description, terms, and symbols.

3.2.5 Logic diagram.

3.2.6 Pin function description.

3.2.7 Schematic circuits. Schematic circuits in terms of the JEDEC Standard No. 12-3, "CMOS Gate Array Macrocell Standard" whenever possible.

3.2.8 Test vectors. NOTE: The test vectors must, as a minimum, detect 95 percent of all detectable single stuck-at-0 and stuck-at-1 faults. (An undetectable fault is defined herein to be a fault for which it is shown that no test or set of tests can detect it. An example would be an unused output of a flip-flop.) The qualifying activity shall approve the particular fault simulator. The vectors shall be supplied on magnetic tape and shall be identified by test tape number and revision letter.

3.2.9 Device electrical performance characteristics. Device electrical performance characteristics shall include dc parametric (see table I herein for minimum requirements), ac functional, and ac critical paths, as selected from manufacturer data sheets. Electrical performance characteristics apply over the full recommended case operating temperature range and specified test load conditions.

3.2.10 Timing diagram. Timing diagram shall show all critical interrelated inputs and outputs. Timing diagrams shall consist of one or more diagrams showing critical interrelationships between two or more signals. Timing requirements such as setup and hold times shall also be shown.

3.2.11 Burn-in circuit. The burn-in circuit shall be as described on figure 1 herein for the AID device.

3.2.12 Maximum power dissipation. Maximum power dissipation shall be in accordance with design.

3.3 Case outlines. Case outlines shall be as specified in MIL-M-38510, appendix C, and 1.2.3 herein.

3.4 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

1/ An altered item drawing (AID), contains all essential information (design, test, etc.) regarding a particular customer personalization of a gate array microcircuit.

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3.4.1 Die attachment material. The use of silver-filled glass die attach material is allowed, subject to requirements as outlined in DESC-EQM letter S4-663, dated 16 May 84, entitled "Silver-Glass Die Attach Requirements MIL-M-38510, FSC 5962."

3.5 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510.

3.6 Electrical performance characteristics. Each manufacturer shall submit to the qualifying activity a data sheet defining the gate array family being qualified. The data sheet shall contain all dc parametric, ac functional and parametric data, and any other data which is manufacturer-dependent but would be considered required by a design engineer. All devices, regardless of manufacturer, must meet the minimum dc parameter limits in table I. All electrical performance characteristics apply over the full recommended case operating temperature range. (The Standard Evaluation Circuit (SEC) (see 4.3.2.2) electrical performance characteristics are as specified in the AID for each manufacturer's SEC.)

3.7 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups in table II herein. The electrical tests for each subgroup are described in the AID (for the user design) and in the AID for each manufacturer's SEC. The electrical test requirements shall, as a minimum follow the requirements of figure 2 and table III herein.

3.8 Marking. Marking shall be in accordance with MIL-M-38510. The AID number shall be added to the marking by the manufacturer. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.8.1 Total dose radiation hardness identifier. The total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and as specified herein.

3.9 Additional line certification requirements. In addition to the requirements of MIL-STD-976, the following documentation shall be provided to the qualifying activity for review and approval. The documentation will remain on file for internal Government use only.

a. Design layout rules as a manufacturer's controlled document.

b. A list of the macros in the manufacturer's macrocell library, macrocell performance data, and macrocell simulation verification data (see 4.3.2.3). A description of the process used by the manufacturer for adding new macrocells to the library shall also be provided.

c. Process control monitor design used by the manufacturer for qualification and wafer acceptance (see 4.3.2.1).

d. Standard evaluation circuit implementation in the form of an AID (see 4.3.2.2 for discussion of this circuit) used by the manufacturer for qualification and quality conformance inspection. (SEC design detail, test vectors, and all required information will be contained in the AID for each manufacturer's SEC.)

e. JEDEC gate array benchmark set (see appendix and 4.3.2.4) delay simulation data.

f. A list of the software packages (including names and current version) used by the manufacturer in the gate array design process.

g. Design rule check (DRC) and electrical rule check (ERC) software verification. The manufacturer's DRC and ERC software shall be run on a design which contains known rule violations; the output shall be presented to the qualifying activity and must show the violations were flagged.

h. Layout versus schematic (LVS) checker. Manufacturer must demonstrate the effectiveness of the LVS software.

i. Fault simulators, used for qualification.

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3.10 Functional delay simulation. To be retained by manufacturer; simulation to be derived from each final application specific electrical design and layout (i.e., post-routed design). Simulation shall be done using actual delays as computed from the placement and layout of the device as it will be fabricated. Actual delays shall include the contribution associated with the delay through the gate, as well as the contribution due to actual metal capacitance and loading on the output(s). Using these actual delays, the application specific design shall insure that there are no timing violations remaining in the circuit. Such timing violations shall include, but not be limited to, setup, hold, critical delay path, and circuit race conditions due to variations in process, temperature, and supply voltage. The simulated circuit behavior at the two (fast and slow) worse case extremes of temperature, supply voltage and process shall be identical states at the specified strobe time (usually at the end of a typical clock cycle where all signals are stable).

3.11 Layout verification. The manufacturer shall retain the results of full, mask level design rule checks, electrical rule checks, and connectivity checks for each application specific design. Rule checking will encompass the rules set provided under 3.9a herein. The manufacturer will explain any rules not checked and all error reports produced by the checker. The LVS check will ensure that the layout matches exactly the logic schematic simulated by the application specific integrated circuit (ASIC) designer.

3.12 Power routing simulation (where available). 2/ To be retained by organization performing the simulation; derived from each final application specific electrical design and layout. The worst case simulation of power busses shall show that at no time shall the localized bus current density exceed specification for allowable current density of the power bus material as defined in MIL-M-38510. In addition, at no point in the power bus shall voltage levels exceed recommended IR drop values from the respective supply. Power routing simulation must be based upon actual placement of cells within the array. Such a simulation may be driven by Monte Carlo methods, or in conjunction with a digital event driven simulator using the selected set of test vectors.

3.13 Procedure for updating certified software packages. Each manufacturer shall submit to the qualifying activity for review the procedure used for approval of updates and revisions of certified software packages (see 3.9f). This procedure shall outline the method and provide test descriptions regarding the process used to accept/reject updates and revisions to in-house or commercially supplied software packages. This requirement only applies to software packages used by the manufacturer in the gate array design process.

3.14 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 123 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein.

4.1.1 Wafer lot acceptance procedure. Each manufacturer shall submit to the qualifying activity for approval, a "Wafer Lot Acceptance Procedure". This procedure shall outline and provide limits regarding the process used to accept or reject a wafer lot. The procedure shall include:

a. Physical measurements of wafer metallization, glassivation and gold backing in accordance with method 5007 of MIL-STD-883.

2/ A manufacturer may take an alternate path for qualification of the specific gate array family members without accounting for specific uses of the array. Such a qualification would be a worst case simulation of each gate array family member by populating all cell positions with the worst case power consumption cells (100 percent utilization). Such a simulation must switch all transistors simultaneously at maximum rated frequency and operating voltage. The contribution to the current density from the output buffers should reflect a TTL load capacitance of 50 picofarads, as well as the TTL dc load current.

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b. Electrical measurements using the process monitor (PM). (NOTE: The PM may be dedicated drop-in or a set of structures in the kerf or scribe line.)

c. Visual inspection to include:

(1) High magnification examination on a sample basis.

(2) Optical test structures to insure alignment, etc.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.10 through 3.1.14 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 1. Ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 1. Ambient temperature (T_A) shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed or included in the final electrical parameter requirements.

e. Constant acceleration shall be in accordance with test condition D, method 2001 of MIL-STD-883.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be 10 percent for static burn-in. Dynamic burn-in is not required, but may be used as an additional screen at manufacturer's option.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits specified in table IV herein, or electrical parameter limits specified in table III, subgroup 2, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection.

4.3.1 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5). In addition, the five-phased qualification outlined below shall be performed through the use of the identified qualification vehicles.

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TABLE I. DC electrical performance characteristics. 1/ 2/

Test	Symbol		Conditions	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	TTL	I_{OH} = -2 mA	2.4		V
		CMOS	$I_{OH} \leq 1$ µA	V_{DD} -0.05		
Low level output voltage	V_{OL}	TTL	I_{OL} = 2.4 mA		0.4	V
		CMOS	$I_{OL} \leq 1$ µA		0.05	
High level input voltage	V_{IH}	TTL		2.0		V
		CMOS		0.7 V_{DD}		
Low level input voltage	V_{IL}	TTL			0.8	V
		CMOS			0.3 V_{DD}	
Input capacitance	C_{IN}		Freq = 1 MHz at 0 V		10	pF
Output capacitance	C_{OUT}		Freq = 1 MHz at 0 V		20	pF
Input curent 3/ 4/	I_{IN}		V_{IN} = V_{DD} or V_{SS}	-1	+1	µA
Three-state output leakage current 3/	I_{OZ}		V_{OUT} = V_{DD} or V_{SS}	-10	+10	µA

1/ This table comprises a core set of performance parameters; all products regardless of manufacturer, shall be within the specified limits.

2/ Limits apply for: $-55°C \leq T_C \leq +125°C$, V_{DD} = 5 V ±10%.

3/ This test applies only to non pull up or pull down devices.

4/ This test limit shall be ±10 µA at -55°C.

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TABLE II. Electrical test requirements.

Line no.	MIL-STD-883 test requirements	Subgroups	
		Class S devices	Class B devices
1	Interim electrical parameters (method 5004)	2 1/	2 1/
2	Static burn-in I (method 1015)	Required	Not required
3	Same as line 1	2/ Plus delta (Δ) limit 2	---
4	Static burn-in II (method 1015)	Required	Required 3/
5	Same as line 1	2/ 3/ Plus (Δ) limit 2*	2/ 3/ Plus (Δ) limit 2*
6	Dynamic burn-in (method 1015)	Required	Not required
7	Same as line 1	Plus (Δ) limit 2/ 2	---
8	Final electrical test parameters (method 5004)	3/ 1,2*,3,7,8, 9,10,11	1/ 3/ 4/ 1,2*,3,7,8, 9,10,11
9	Group A test requirements (method 5005)	5/ 1,2,3,4,7,8, 9,10,11	1/ 3/ 5/ 1,2,3,4,7,8, 9,10,11
10	Group B end-point electrical parameters (method 5005)	2/ 6/ 1,2,3,9, 10,11 Plus (Δ) limit	N/A
11	Group C end-point electrical parameters (method 5005)	N/A	1/ 2/ 1, 2 Plus (Δ) limit
12	Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2

1/ Subgroups 1, 2, 3, 7, and 8 form a part of subgroups 9, 10, and 11 tests.

2/ (Δ) indicates a delta limit shall be required only on table III, subgroup 2, and the delta values shall be computed with reference to the previous interim electrical parameters (line 1).

3/ (*) PDA applies to subgroup 2 (see 4.2.1).

4/ The device manufacturer may, at his option, either complete the required electrical parameter measurements (subgroup 2), including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete the required electrical measurements (subgroup 2) without delta measurements within 24 hours after burn-in completion (removal of bias).

5/ Capacitance testing (see 4.4.1d).

6/ Electrostatic discharge sensitivity and latch-up tests require subgroup 1 only.

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TABLE III. Group A inspection for all device types.

Subgroup	Test	Symbol	Test no.	MIL-STD-883 method	Conditions	Measured terminal	Limits
T_C = +25°C 1/	Input clamp voltage, positive	V_{IC}, pos	1	---	V_{DD} = GND; 1 mA into terminal	All input terminals	Per manufacturer's data.
	Input clamp voltage, negative	V_{IC}, neg	2	---	GND = V_{SS}; -1 mA into terminal	All input terminals	Per manufacturer's data
	Output voltage high, TTL	V_{OH}, TTL	3	3006	V_{DD} = 4.5 V	All applicable output terminals	See table I
	Output voltage high, CMOS	V_{OH}, CMOS	4	3006	V_{DD} = 4.5 V	"	"
	Output voltage low, TTL	V_{OL}, TTL	5	3007	V_{DD} = 5.5 V	"	"
	Output voltage low, CMOS	V_{OL}, CMOS	6	3007	V_{DD} = 5.5 V	"	"
	Input current, high	I_{IH}	7	3010	V_{DD} = 5.5 V; input voltage = 5.5 V	All input terminals	"
	Input current, low	I_{IL}	8	3009	V_{DD} = 5.5 V; input voltage = V_{SS}	All input terminals	"
	Static supply current	I_{DDS}	9	3005	V_{DD} = 5.5 V	V_{DD} terminals	Per design
T_C = +125°C 2 1/	Same terminal conditions and limits as above, except T_C = +125°C.						
T_C = -55°C 3 1/	Same terminal conditions and limits, except T_C = -55°C.						
T_C = +25°C 4 1/	Input capacitance, negative	C_{IN}	10	3012	F = 1 MHz	All input terminals	See table I
	Output capacitance high, TTL	C_{OUT}	11	3012	F = 1 MHz	All output terminals	See table I

See footnotes at end of table.

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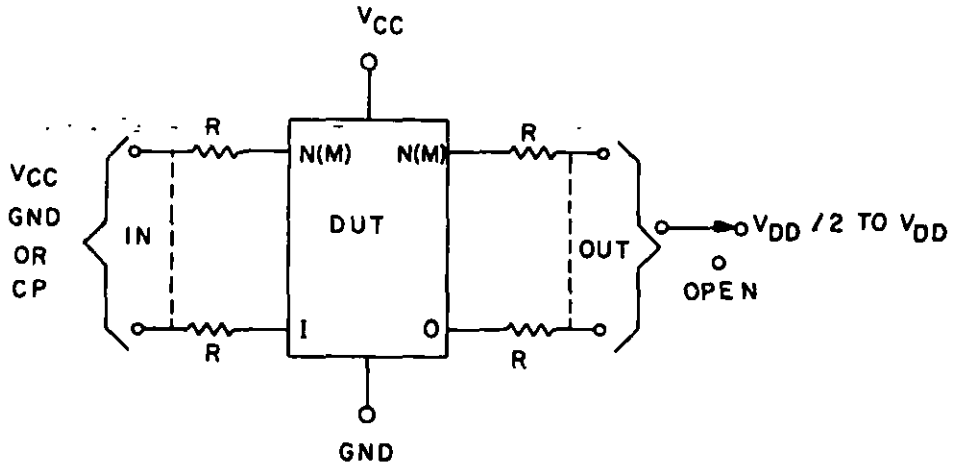
TABLE III. Group A inspection for all device types - Continued.

Subgroup	Test	Symbol	Test no.	MIL-STD-883 method	Conditions	Measured terminal	Limits
7 8	Functional	---	See fig. 2	3014	V_{DD} = 4.5 V	Design specific; according to AID	
9 2/ T_C = +25°C	AC switching	---	See fig. 2	3003	V_{DD} = 4.5 V	Design specific; according to AID	
10 2/ T_C = +125°C	Same as subgroup 9, except T_C = +125°C.						
11 2/ T_C = -55°C	Same as subgroup 9, except T_C = -55°C						

1/ Static tests shall be conducted as required by table II of this specification; limits and conditions appear in table I herein. Particular terminal(s) to be measured are design specific and must appear in the applicable AID.

2/ Switching tests, comprising subgroups 9, 10, and 11, shall be conducted as required by table II. The test vectors to accomplish the functional and switching verfication are design specific, and must be included in the AID. The vector set must meet the criteria as outlined in 3.2.8. Limits appear in the AID.

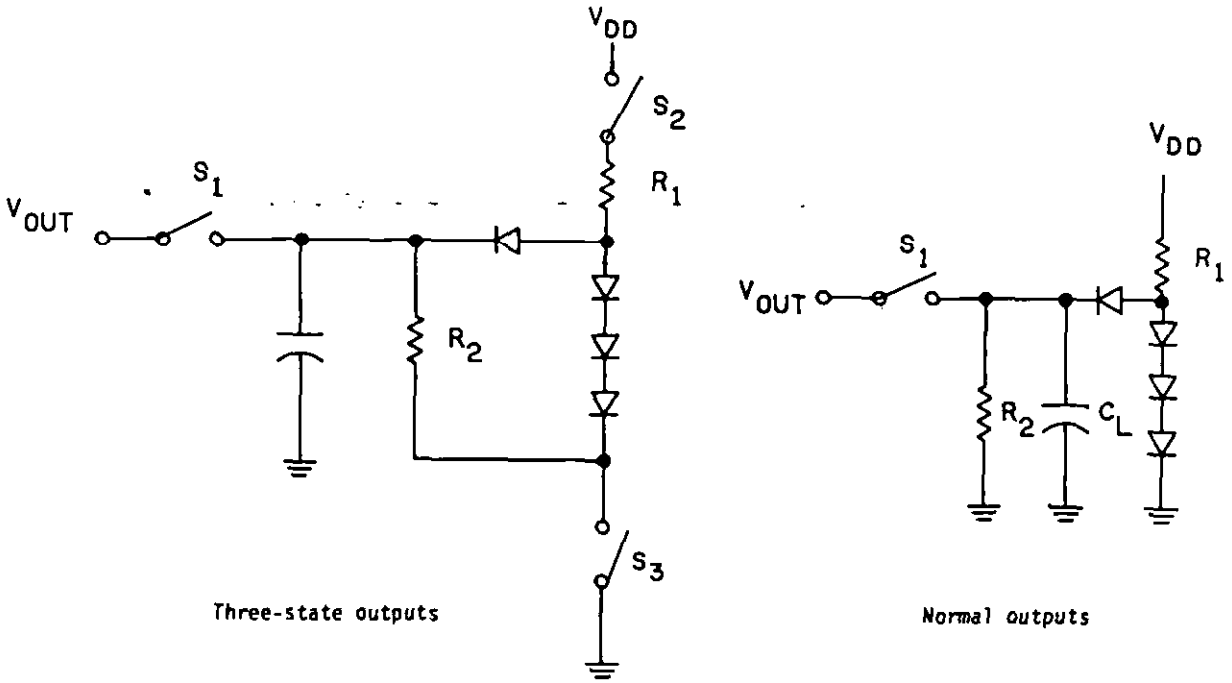
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NOTES:
1. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{DD}/2$ to V_{DD}. Resistors are optional on outputs if open. Resistors are required on inputs and outputs connected to V_{DD}. R = 680Ω to 47 kΩ, actual resistor value shall reflect worst case load conditions.
2. For static burn-in II, all inputs shall be connected through resistors to V_{DD}. Outputs shall be open or connected to $V_{DD}/2$ to V_{DD}. Resistors are optional on outputs if open. Resistors are required on inputs and on outputs connected to $V_{DD}/2$ to V_{DD}. R = 680Ω to 47 kΩ, actual resistor value shall reflect worst case load conditions.
3. For dynamic burn-in, all inputs shall be connected through the resistors in parallel to a common CP. Outputs shall be connected to $V_{DD}/2$ ±0.5 V through the resistors. R = 1 kΩ ±5% for outputs, 680Ω to 47 kΩ for inputs.
4. CP = 100 kHz ±50%; duty cycle = 50 ±15%; V_{IH} = 4.5 V to V_{DD}; V_{IL} = 0 ±.5 V, transition time ≤ 0.5 µs.
5. V_{DD} = 5.0 V ±0.5 V.
6. I = input, O = output, N (M) = highest numbered input (output).
7. In the case of multiple V_{DD} or V_{SS} pins, all connections shall be tied to the appropriate level.
8. Pin designations determined by customer's configuration.

FIGURE 1. Burn-in circuit (device specific).

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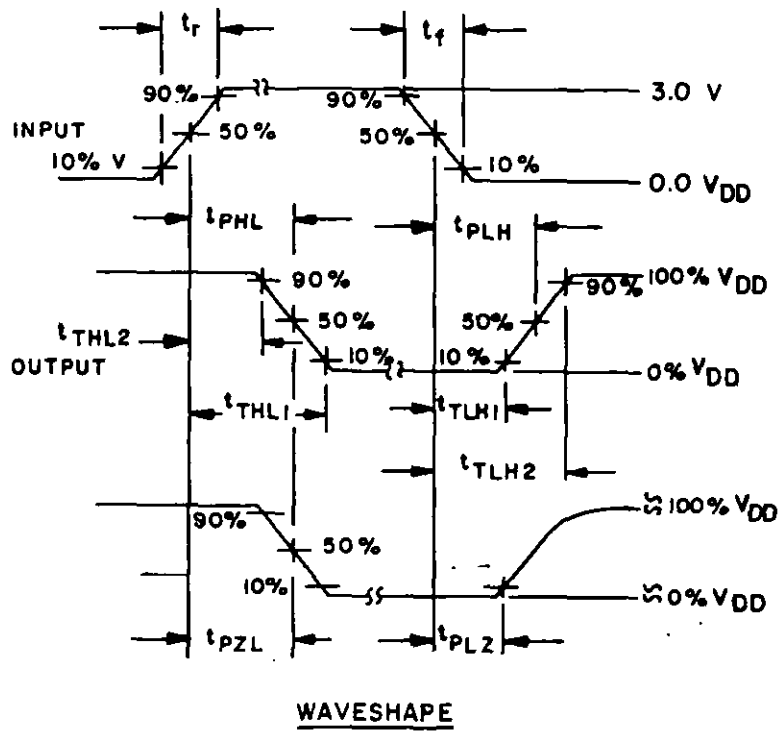


FIGURE 2. Switching time test circuit and waveforms.

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NOTES:
1. C_L is determined by the specific ATE (automatic test equipment) being utilized, plus jig and probe capacitance. The total load will be specified in the AID and all switching test limits will be based on the stated load.
2. Input pulse shall have the following characteristics: $t_r = t_f \leq 6$ ns; PRR $\leq$ 1 MHz; duty cycle = 50%.
3. All unused inputs are tied either to V_{DD} to GND.
4. $t_{THL1} = t_{THL2} = t_{THL}$; $t_{TLH2} = t_{TLH1} = t_{TLH}$.
5. S_1 is closed during function test and all ac tests, except output enable tests.
6. S_1 and S_3 are closed while S_2 is open for t_{PZH} test.
   S_1 and S_2 are closed while S_3 is open for t_{PZL} test.
7. R_2 = 1 kΩ for three-state output.
   R_2 is determined by I_{OH} at V_{OH} ±2.4 V for non three-state outputs.
8. R_1 is determined by I_{OL} (min) with V_{DD} -.05 V minus the current to ground through R_2.
9. Switching tests shall be performed utilizing load circuits shown, or the equivalent.

FIGURE 2. Switching time test circuit and waveforms - Continued.

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4.3.2 Qualification phase and vehicles.

Qualification phase	Qualification vehicles
Control and stability	Process monitor die (PM)
Process reliability	Standard evaluation circuit (SEC)
Macrocell design and simulation	Macro test chip set
CAD routing and post routing simulation	JEDEC standard for gate array benchmark set
Design check software verification	Manufacturer test case

NOTE: Qualification of the macrocell design and simulation, and CAD and post routing simulation will be based on the manufacturer's ability to design, place and route, and manufacture devices where actual measured performance characteristics (i.e., propagation delays, rise and fall times, drive characterization, trigger levels etc.) are within simulated device performance limits (see 4.3.2.4).

4.3.2.1 Process control and stability. Process control and stability of dc parameters must be demonstrated through the use of the manufacturer's process monitor (PM). The PM (either a dedicated drop-in or structures fabricated in the kerf) shall be designed so that the dc process parameters (ac parameters may be included as a manufacturer's option) may be measured in wafer form or packaged device form. The PM design must be submitted to the qualifying activity for approval prior to use for qualification and must contain, as a minimum, the following structures:

a. N channel device (minimum geometry).

b. P channel device (minimum geometry).

c. N channel device (large device).

d. P channel device (large device).

e. Sheet resistance measurement structure.

f. Metal step coverage structure.

g. Field threshold device(s).

h. Intermetal oxide integrity structure.

i. Contact chains (to be sufficient length to be representative of the contact resistance).

(1) Metal to poly (where applicable).

(2) Metal 1 to metal 2 (where applicable), via resistance.

(3) Gated inverter chain(s) (for ac measurement; may be included as a manufacturer's option).

(4) Metal to diffusion (where applicable).

For qualification, PM's on a minimum of 3 different lots (minimum of 4 PM's per wafer) shall be measured to insure the establishment of a statistically valid data base on which a decision can be made as to whether the manufacturer's process is stable and under control. Recent lot history data can be used to satisfy this requirement (at the manufacturer's option). (For quality conformance inspection, measurement of PM parameters will be required for wafer lot acceptance (see 4.1.1).)

4.3.2.2 Process reliability. The process reliability is to be qualified using the manufacturer's standard evaluation circuit, referred to as the SEC. The SEC design shall be submitted to the qualifying activity for approval prior to use in the form of a manufacturer's altered item drawing, and as such shall contain the basic information as detailed in sections 3.2.1 through 3.2.10. It shall be fabricated with the same process that will produce any application specific gate array device under this detail specification. The SEC design shall be configured in such a manner

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so as to evaluate the reliability of the underlayer designs (diffusions, etc.) and evaluate worst case design rule conditions on the personalization layers. The design should utilize library macrocells to form oscillator rings and also include test structures which will detect metal-to-metal shorting or opening, high via resistance and dielectric pinholes during reliability life testing. The SEC shall be implemented on the largest member (i.e., number device type with the largest number of gates and if applicable with the highest pin count package of the gate array family. Smaller gate count members of a manufacturer's gate array family in the highest pin count package or in smaller pin count packages may be qualified by extension (see 4.3.3).

The SEC will also be the main qualification vehicle for the electrical testing which follows (4.4.1, 4.4.2, 4.4.3, and 4.4.4). Test limits appear in the AID (see table I herein for minimum dc parameters) for each manufacturer's SEC. The electrical tests for each subgroup shall be described in table III. The SEC shall be suitable for static bias aging, as life testing will be performed on this device (see 4.4.3).

4.3.2.3 Macrocell design and simulation qualification. The macrocell design and simulation qualification shall be accomplished in a two step procedure consisting of parameter verification or simulation and functional verification.

A chip shall be designed to provide access to a set of macrocells to test performance characteristics. These macrocells are a subset of the JEDEC Standard No. 12-3, "CMOS Gate Array Macrocell Standard"; macrocell substitutions are allowable only if the manufacturer does not offer a particular macrocell, and such substitutions must be approved by the qualifying activity. The set of macrocells shall include:

Macrocell (*)	Description
JIV1	Inverter
JND4	4-input NAND
JAO12	2-input AND into 3-input NOR
JLDR1	D latch with active low reset
JFJKR1	JK flip-flop with active low reset
JBIT1	TTL input buffer
JBIC1	CMOS input buffer
JBO1	Output buffer
JBTB2	3-state I/O buffer with pull up

(*) JEDEC naming convention

The intent is to get a representative cross section of macrocell types (i.e., combinational, sequential, input, output). Chains shall be formed (when necessary to avoid rise and fall time measurement problems) and actual performance data over the full operating range shall be taken (a provision to extract for multiplexing and I/O buffer delay shall be included). Delay versus metal wire length and fanout for the above macrocells shall be determined. The actual performance data shall be submitted to the qualifying activity along with computer program simulation results. The actual performance data must be within the limits predicted by the simulation. If multipliers are used to extrapolate performance at the temperature extremes, such multipliers shall be verified as well.

In addition, for the above macrocells, a set of pins shall be provided on the test chip for observability. This will enable verification of functionality of the macrocells. (NOTE: Inputs and outputs may be multiplexed.)

4.3.2.4 CAD routing and post routing simulation. A chip or set of chips incorporating JEDEC Standard No. 12, "Standard for Gate Array Benchmark Set" (see appendix herein) shall be used to qualify the manufacturer's ability to perform routing and to accurately predict post routing performance. The manufacturer must submit to the qualifying activity:

a. The actual measured performance data for each function in the benchmark over temperature and voltage.

b. The computer simulation performance prediction. The two results will remain on file and the actual measured performance must fall between the two worst case (i.e., fast worse case V_{DD} = 5.5 V at -55°C and slow worse case V_{DD} = 4.5 V at +125°C) simulation performance prediction limits.

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4.3.2.5 Design check software verification. The manufacturer shall verify that the DRC, ERC, and LVS software is capable of performing the intended function. This shall be accomplished by running a design with known errors against the design check software and demonstrating that the errors were caught.

4.3.3 Microcircuit qualification extension. For qualification extension if a manufacturer part I qualifies the SEC (i.e., the highest gate count array) in the highest pin count package type, then lower gate count arrays in that package type that are manufactured identically (i.e., same line, same process) as the SEC may be part I qualified upon the approval of the qualifying activity.

a. Lower gate count arrays in lower pin count packages that are manufactured identically (i.e., same line, same process) as the SEC may be part I qualified upon approval of the qualifying activity using another die provided the gate utilization is 60 percent or greater and by conducting group A electrical tests and any electrical subgroups added to group C tests in table II herein for the die extension.

b. Packages extension, if required by qualifying activity on lower pin count packages, may be achieved by performing subgroups 2, 3, 4, 5, and 7 of group B tests and all group D tests of method 5005 of MIL-STD-883. Data must be submitted in accordance with MIL-M-38510, appendix D.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test requirements shall be as specified in table II herein and as follows.

b. The vehicles to be used for the group A inspection are as follows:

(1) Qualification - standard evaluation circuit.

(2) Quality conformance inspection - actual user device (via the altered item drawing).

c. Subgroups 5 and 6 of table I, method 5005 of MIL-STD-883 shall be omitted.

d. Subgroup 4 (C_{IN} and C_{OUT} measurements) shall be measured only for initial qualification and after process or design changes which may affect the value. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz. One pin of each input/output driver (buffer) type shall be tested on each sample device.

e. Dynamic and functional testing shall be performed using the circuit configuration shown on figure 2 herein or equivalent.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II, method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein, and as follows:

a. The vehicles to be used for the group B inspection are as follows:

(1) Qualification - standard evaluation circuit.

(2) Quality conformance inspection - actual user device (via the altered item drawing).

b. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspection and shall consist of tests specified in table IV herein.

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c. A latch-up test shall be performed on a special subgroup using an LTPD of 15. One of each input and output configuration (i.e., buffer) of the device under test (DUT) shall be subjected to a current pulse of ±150 mA amplitude and 500 ms duration under the following conditions:

(1) V_{DD} = +5.5 V, V_{SS} = GND.

(2) T_A = +125°C.

The selected pin will be subjected to the current pulse, while I_{DD} is monitored for latch-up. Untested inputs shall be tied to $+V_{DD}$. Untested outputs shall be open. The device shall pass all electrical tests (subgroup 1) after latch-up evaluations have been performed.

d. Latch-up sensitivity and ESD tests shall be performed on a special subgroup for qualification and after input or output buffer design changes.

e. Electrostatic discharge sensitivity testing shall be in accordance with MIL-STD-883, method 3015, except that each pin combination shall be required on only one pin of each input and output driver (buffer) type.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III, method 5005 of MIL-STD-883 and as follows:

a. Qualification and quality conformance inspection shall be performed on the manufacturer's SEC.

b. End-point electrical parameters shall be as specified in table II with limits shown on the AID covering the manufacturer's SEC (see table I herein for minimum dc parameters).

c. Steady-state life test (method 1005 of MIL-STD-883).

(1) Use life-test condition A using the SEC and the circuit configuration shown on figure 1. The life test circuit schematic shall be submitted to the qualifying activity for approval prior to use.

(2) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

(3) T_A = +125°C minimum.

d. The following sampling plan shall be used in lieu of table III of method 5005 for all group C subgroups: A sample of 10 SEC devices shall be chosen and submitted to test with no failures allowed. If not more than 1 failure is found in the first sample of 10, a second sample of 10 SEC devices is permitted with no further failures allowed.

e. End-point electrical parameters for class B, life test shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

f. Constant acceleration shall be performed using test condition D.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV, method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein. The SEC shall be used for group D inspection whenever practical; for smaller packages where the SEC cannot be used, another die may be used provided the gate utilization is 60 percent or greater. Group D inspection shall be performed in accordance with MIL-M-38510, section 4.5, except as defined below:

a. If more than one package type within a case outline family is qualified, then the group D inspection shall be rotated between the qualified package types. A different package type shall be tested every three months based on the inspection lot date code.

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b. If a package type is skipped in the rotation due to the absence of production, then the first lot of the skipped type shall receive group D inspection. The successful completion of this package type shall establish a new three-month period before the next package type in the rotation is selected for group D inspection.

c. Failure of a group D subgroup shall discontinue coverage of all package types within the case outline family. Lot-by-lot inspection shall be initiated until three consecutive lots pass. The package type which failed the subgroup shall not be shippable until that package type has successfully passed the previously failed subgroup.

d. Constant acceleration shall be performed using test condition D.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.8.1). When group E testing is performed, it shall be in accordance with table V of method 5005 of MIL-STD-883.

TABLE IV. Delta limits at +125°C.

Parameter 1/	Device types
	All
I_{DD}	±10% or 20 µA whichever is greater

1/ The above parameter shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltage values given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

4.5.3 Quiescent supply current (I_{DD} test). When performing quiescent supply current measurements (I_{DD}), the meter shall be placed so that all currents flow through the meter.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510. In addition, wherever practical, all devices shall be in contact with a conductive material which shorts all leads together to prevent electrostatic damage.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment), design applications, and logistic purposes.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents shall specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

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c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis, including required test condition of method 5003 of MIL-STD-883, corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable.

h. Requirements for "JAN" marking.

i. Requirements for altered item drawing provided to the manufacturer; see 3.2.

j. Requirements for total dose radiation testing (see 3.8.1 and 4.4.5), if applicable.

6.3 Logistic support. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.5). Longer length leads and lead forming shall not affect the part number.

6.4 Handling. MOS devices must be handled with certain precautions to avoid damage due to the discharge of accumulated static charge. These CMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage; however, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive foam or carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_{IN}	Input terminal-to-GND capacitance.
GND	Ground zero voltage potential.
I_{DD}	Quiescent supply current.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.

6.6 Manufacturers' designations. Manufacturers of device types on this specification are designated as shown in table V herein. This does not imply qualification or certification, refer to QPL-38510 for information regarding qualified sources.

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TABLE V. Manufacturers' designations.

Device type	A	B	C	D	E	F	G	H
	Mostek/ Thompson	UTMC	GE	LSI	Motorola	TI	Siliconix	National

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Manufacturer specific generic-industry type
01 through 17	See listed manufacturer (6.6 and table V) data sheets

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APPENDIX

GATE ARRAY BENCHMARK SET

10. SCOPE

10.1 Scope. This appendix contains benchmark description numbers from JEDEC Standard No. 12, "Standard for Gate Array Benchmark Set" as required by 4.3.2.4 CAD routing and post routing simulation herein.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Benchmark number description.

1 4-bit ALU, similar to 54S381

2 16-bit ALU, similar to 54S381

3 4-bit rotator

4 16-bit rotator

5 8-bit register

6 8-bit up/down counter

7 3- to 8- decoder, similar to 54S138

8 9-bit parity generator, similar to 54S280

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CONCLUDING MATERIAL

Custodians:
Air Force - 17
Navy - EC
Army - ER

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1034-1)

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